

COEN ADLER

☎ 510-314-9393 ✉ ctadler@uci.edu [in coen-adler](#) [Coaster41](#) [Website](#) 🏠 Irvine, CA

Education

University of California, Irvine

Ph.D. in Computer Science - 06/29

September 2024 – June 2029

GPA: 3.97

University of California, Santa Cruz

Bachelor of Science in Computer Science

September 2021 – June 2024

GPA: 3.94

Work Experience

Cisco SEA

Software Engineer Intern II

June 2024 – September 2024

San Jose, CA

- Architected an extensible Golang asset connection workflow engine for batch command execution and response analysis
- Processed unstructured device health and security data using Splunk and Ollama to visualize fleet safety
- Reduced batch computing overhead from Apache Guacamole by 80% through multi-threaded integration in REST API server and SSH client engine

UCSC-ACM AI Research Lab

Co-Founder & Research Advisor

March 2023 – June 2024

Santa Cruz, CA

- Founded an artificial intelligence research group aiming to introduce academic research to undergraduate students
- Advised and conducted novel research on various machine learning subjects including map generation using Variational Autoencoder (VAE) U-Nets, aerodynamic simulations using Physics Inspired Neural Networks (PINNs), and applications of Meta's Segment Anything Model (SAM) on medical imaging
- Designed and taught an accredited course guiding students on research fundamentals and introduction to deep learning models and techniques

Cisco IoT

Software Engineer Intern I

July 2023 – September 2023

San Jose, CA

- Implemented troubleshooting feature, application manager microservice on Cloud IoT Operations Dashboard providing an accessible platform for non-developers to help service cloud devices
- Developed a robust templating service using Apache Freemarker allowing for runtime workflow modification
- Utilized Java, PostgreSQL, and SpringBoot to develop a scalable REST API to be used in tens of thousands of edge devices and native applications

Cisco Webex

Software Engineer Intern I

July 2022 – September 2022

San Jose, CA

- Utilized Kubernetes and Docker to transition to AWS Spot EC2 Instances reducing resource costs by up to 90%
- Prototyped automatic node shutdown, cluster scaling, and Helm-based app installer reducing 70% of unused node costs
- Facilitated code stack restructure from x86 compute units to AWS Graviton ARM resulting in 50% cost reductions

Publications

Beyond Accuracy: Are Time Series Foundation Models Well-Calibrated? | *ICLR Conference*

April 2026

- First-authored conference paper analyzing the effects of various TSFM components on forecasting calibration
- Trained and evaluated SOTA TSFMs varying prediction heads, comparing quantile, single-distribution, and mixture-distribution heads
- Developed modular evaluation pipeline to analyze autoregressive generation variations

Calibration Properties of Time Series Foundation Models: An Empirical Analysis | *ICML Workshop*

August 2025

- First-authored workshop paper analyzing systematic calibration and over-/under-confidence of TSFMs
- Designed and compared custom TSFM projection heads and autoregressive long-term forecasting methods

Model Pruning and Explainability: An Experience Report | *Undergraduate Thesis*

August 2024

- Assessed the effects of neural network pruning on the explainability of computer vision models
- Applied and re-implemented a variety of pruning methods on VGG and Resnet models which showed a large degradation in object detection and image classification

Fair Augmentation of Decision Trees Through Selective Node Retraining | *Baylearn Symposium*

October 2023

- First-authored paper on fair retraining of decision trees for efficient bias removal with minimal effects on accuracy
- Designed node selection and retraining method reducing training time by 50% compared to full tree reformulation
- Invited to present paper abstract at Baylearn Symposium to an audience including Google, Apple, Netflix, AT&T

Diffusion Models for Species-Specific Functional Habitat Connectivity | *Preprint - Under Review* December 2023

- Developed diffusion model that utilizes GPU parallelization for mapping bird habitat connectivity sponsored by Google
- Implemented back propagation enabling for generation of a flow map over 100 times faster than existing methods
- Invited to present paper at Google’s Geo For Good Conference for work in accessible species conservation efforts

Utilizing SAM For Assessing Localization of GRAD-CAM in Medical Imaging | *arXiv* June 2023

- Advised a team of five to develop a novel method for increasing accuracy of medical image segmentation by 30%
- Applied Meta’s Segment Anything Model (SAM) on automated bounding box generation using GRAD-CAM to accurately segment Malaria infected cells

Current Research

Mechanistic Interpretability in Time Series Foundation Models | *Pytorch, TransformerLens* July 2025 – Present

- Evaluating the impact of architectural decisions on performance or forecast style from a mechanistic interpretability lens
- Exploration of feature explanations and Superposition in Time Series Foundation Models using Sparse Autoencoders

Google - Time Series Anomaly Detection | *Darts, StatsForecast, Python* December 2024 – Present

- Collaboration with Google for time series anomaly detection evaluating classical and foundation models on univariate large-scale datasets
- Explanations for detected anomaly severity to reduce impact of false positives via XAI methods including SHAP, Integrated Gradients, and GradCAM

Tabular Foundation Models Exploration October 2024 – Present

- Analysis on existing foundation models for tabular data, comparing transformer and finetuned LLM models

Awards and Fellowships

UC Irvine Computer Science Department Research Fellowship | *UC Irvine* 2024

Chancellor’s Undergraduate Research Award | *UC Santa Cruz* 2024

Deans’s Undergraduate Research Award | *UC Santa Cruz* 2024

Dean’s Award | *UC Santa Cruz* 2021-2024

Technical Skills

Languages: Python, Golang, Java, C/C++, SQL, R, Flutter/Dart
Tools: Git, Kubernetes, Slurm, Docker, PostgreSQL, OpenAPI, GCP, AWS, Gradle, Helm, Neo4j
Frameworks: PyTorch, TensorFlow, SpringBoot, Pandas, Numpy, Ollama, Scikit-learn